Lab No. 5

Build, Test, and Package Applications with IntelliJ IDEA and Docker

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REGISTRATION NO.: 2023-BCS-053

COURSE CODE: CS-649

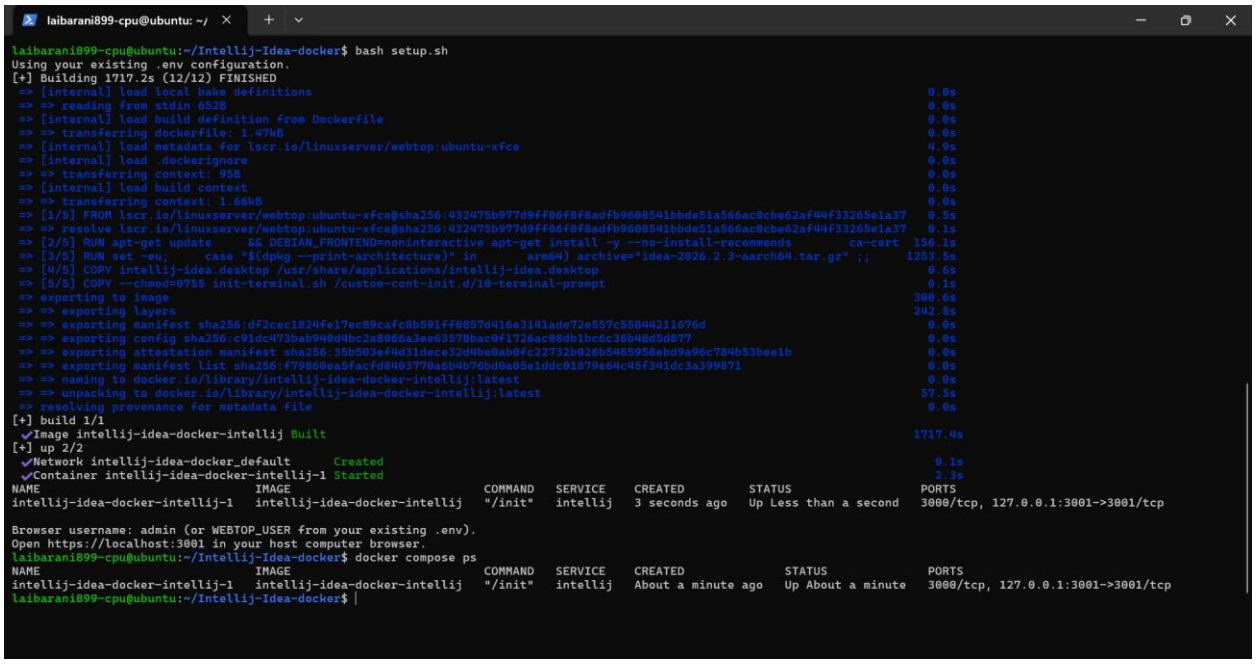
SECTION: A

GITHUB USERNAME: laibarani899-cpu

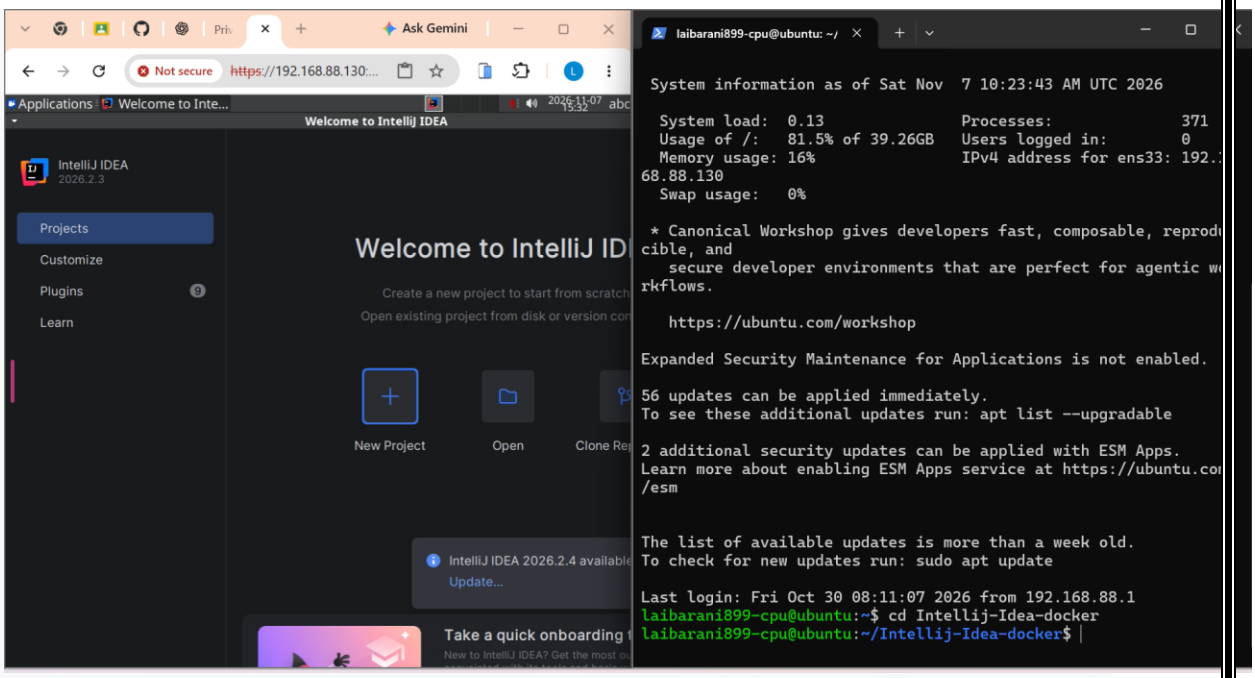
Task 1: Clone and Launch IntelliJ IDEA

```
laibarani899-cpu@ubuntu:~$ cd ~
laibarani899-cpu@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Intellij-Idea-docker.git
Cloning into 'Intellij-Idea-docker'...
remote: Enumerating objects: 35, done.
remote: Counting objects: 100% (35/35), done.
remote: Compressing objects: 100% (32/32), done.
remote: Total 35 (delta 16), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (35/35), 15.24 KiB | 1.09 MiB/s, done.
Resolving deltas: 100% (16/16), done.
laibarani899-cpu@ubuntu:~$ cd ~/Intellij-Idea-docker
laibarani899-cpu@ubuntu:~/Intellij-Idea-docker$ ls
Dockerfile  README.md  compose.yaml  init-terminal.sh  intellij-idea.desktop  remove.sh  resources.sh  setup.sh
laibarani899-cpu@ubuntu:~/Intellij-Idea-docker$
```

In **Ubuntu host terminal**, clone the supplied repository:

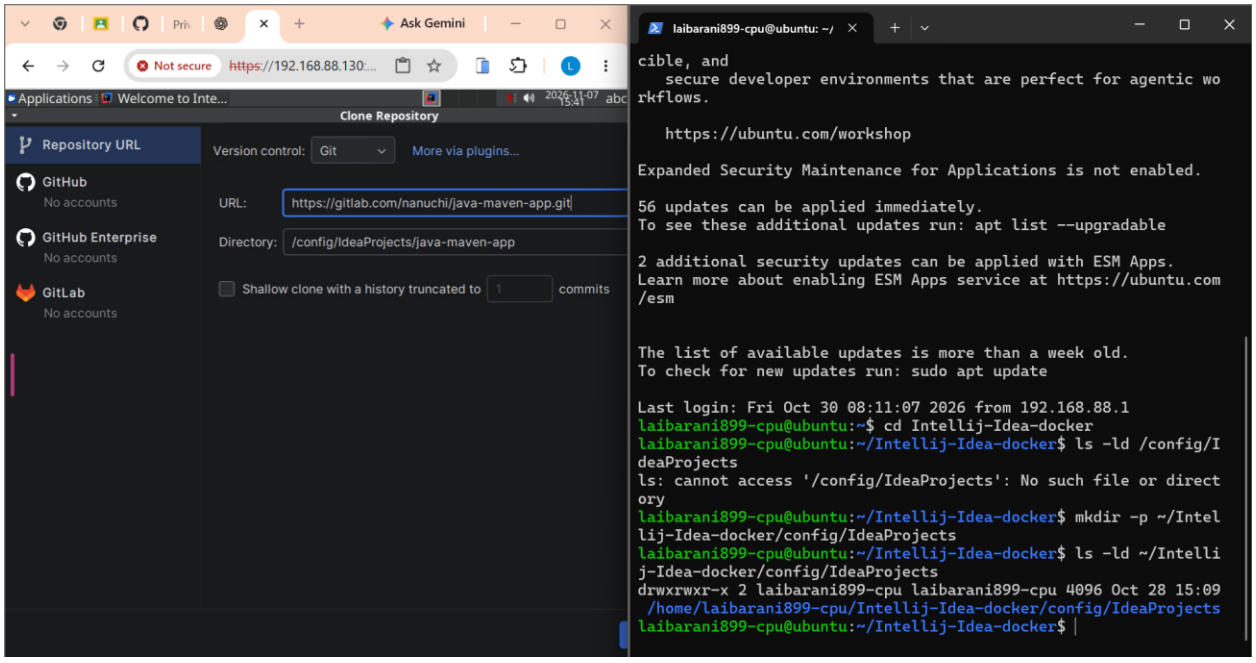


Build and launch the environment



Enter and confirm the browser password when prompted. The script builds the image and starts it with Docker Compose. The first build downloads the desktop and IDE.

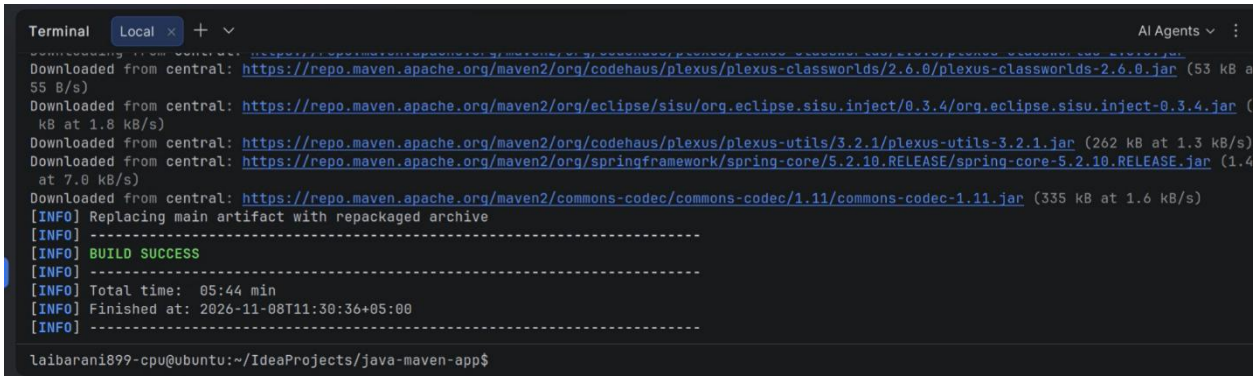
Task 2: Build and Run the Maven Project



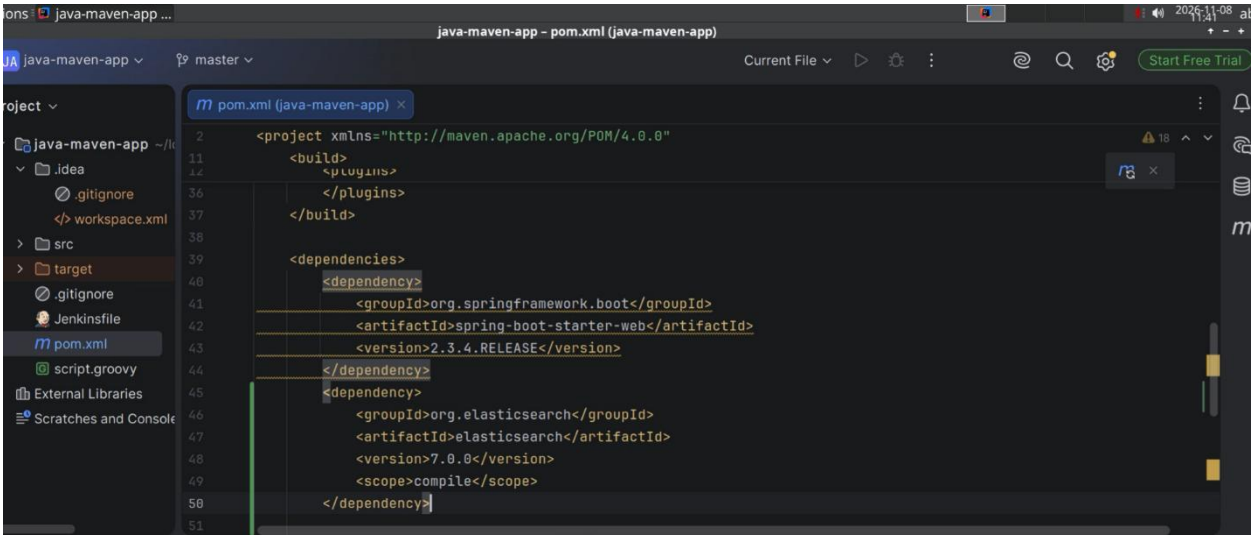
On IntelliJ's Welcome screen, select **Clone Repository** or **Get from Version Control**.



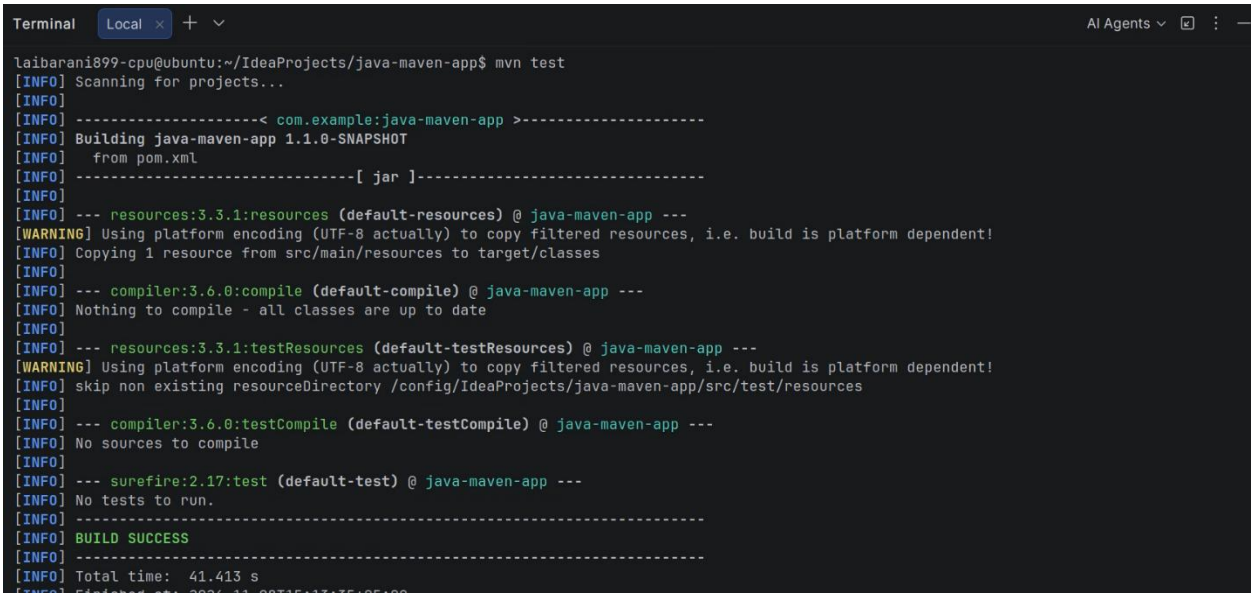
Open **View** → **Tool Windows** → **Terminal** inside IntelliJ. Work from the Maven project's root directory.



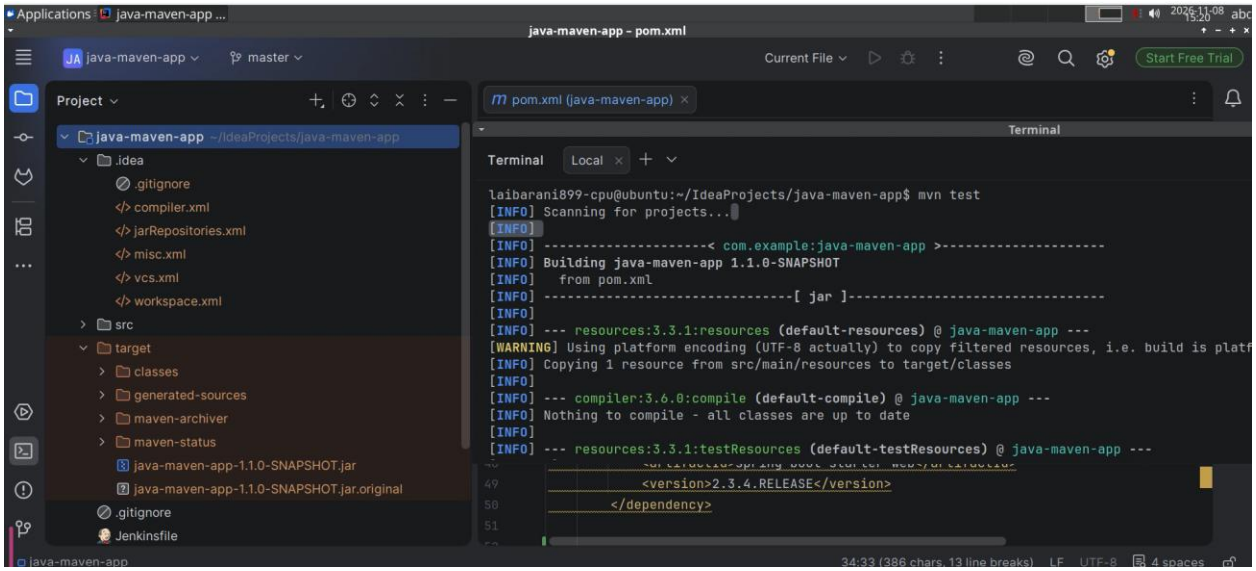
Package the application



Open **`pom.xml`** in IntelliJ. Add this dependency inside the existing **`<dependencies>`** section, alongside the other project dependencies:



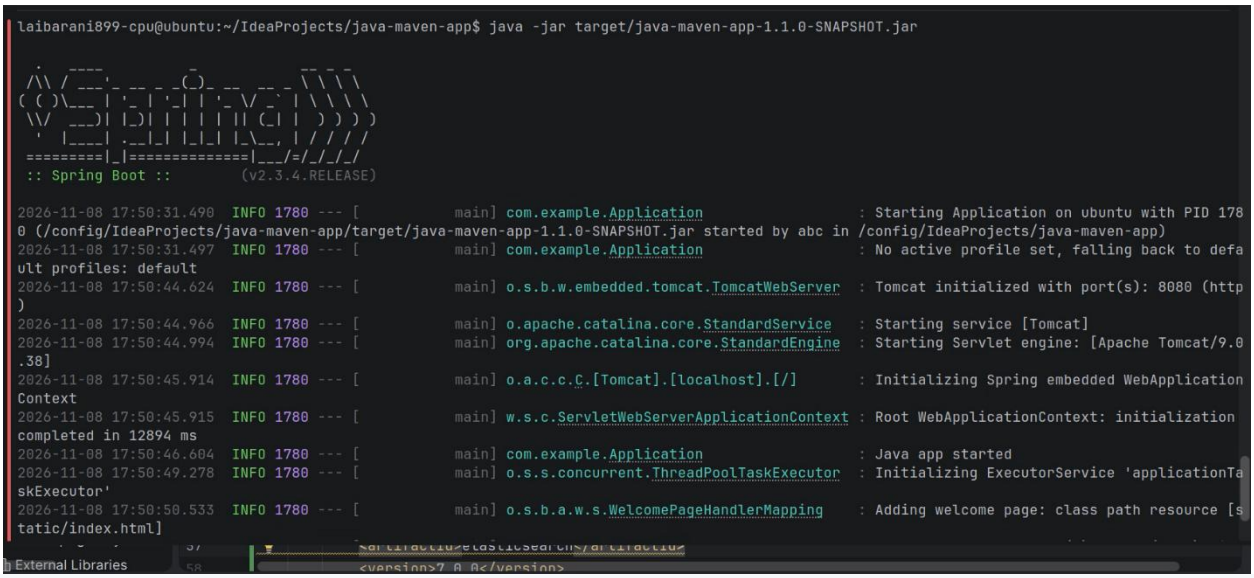
Run Maven's test phase:



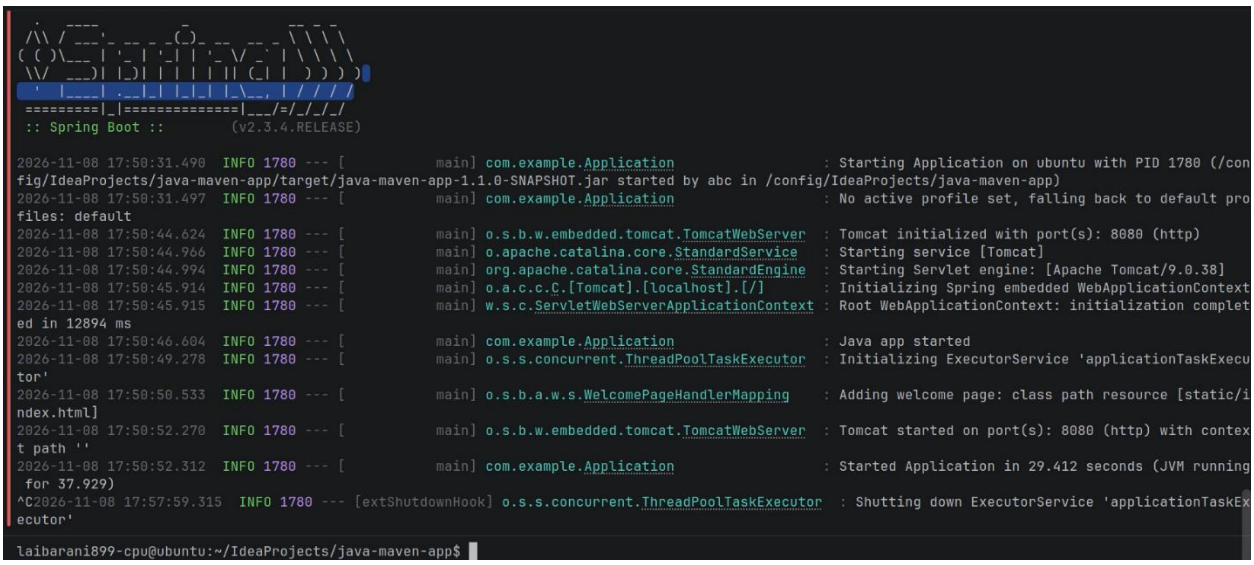
In IntelliJ's Project pane, refresh the project if needed and expand **target**. Locate

```
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
<version>2.3.5.RELEASE</version>
<configuration>
    <mainClass>com.example.Application</mainClass>
</configuration>

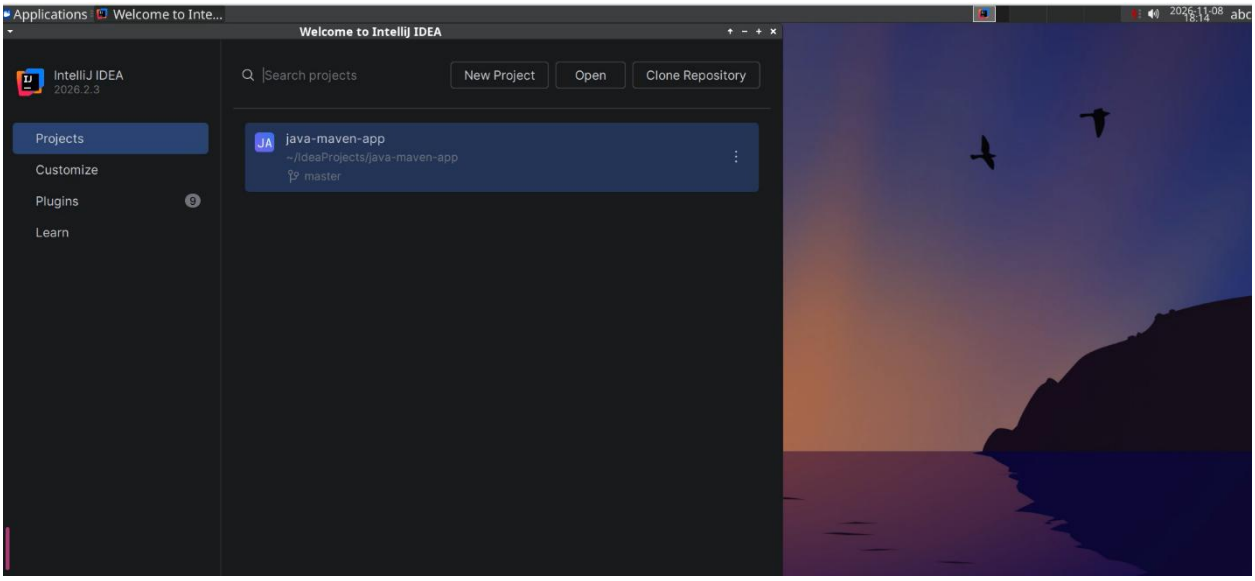
<executions>
    <execution>
        <goals>
            <goal>repackage</goal>
        </goals>
    </execution>
</executions>
```



showing the JAR command and successful application startup.

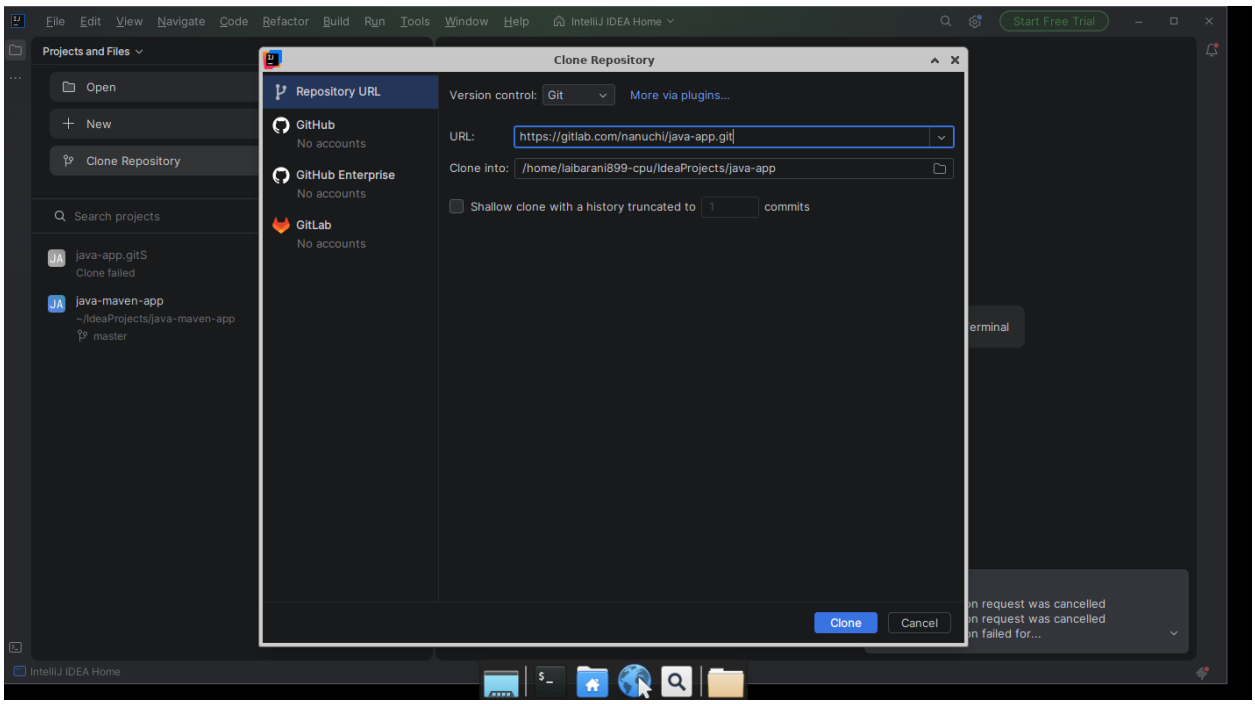


Press **Ctrl+C** in that terminal to stop the application. Wait for the shell prompt to return

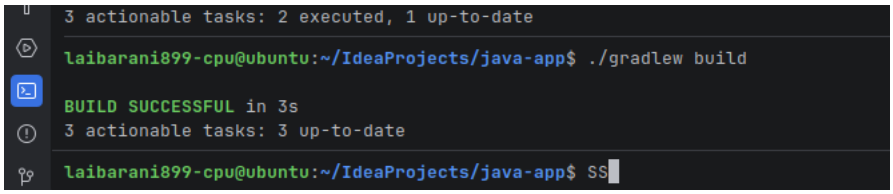


Select **File** → **Close Project** and return to IntelliJ's Welcome screen.

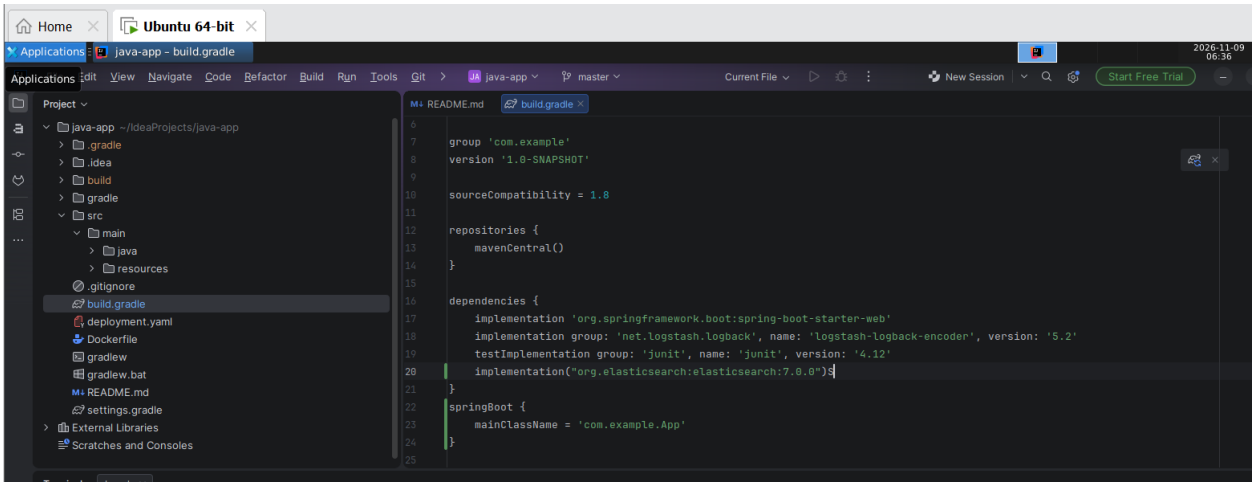
Task 3: Build and Run the Gradle Project



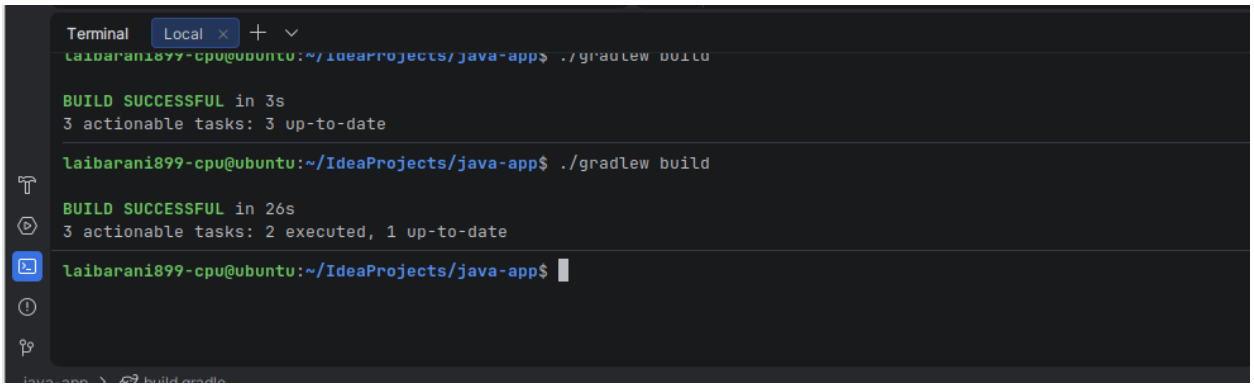
Click **Clone**, open/trust the sample and load it as a Gradle project. It contains `build.gradle`, `settings.gradle` and `gradlew`.



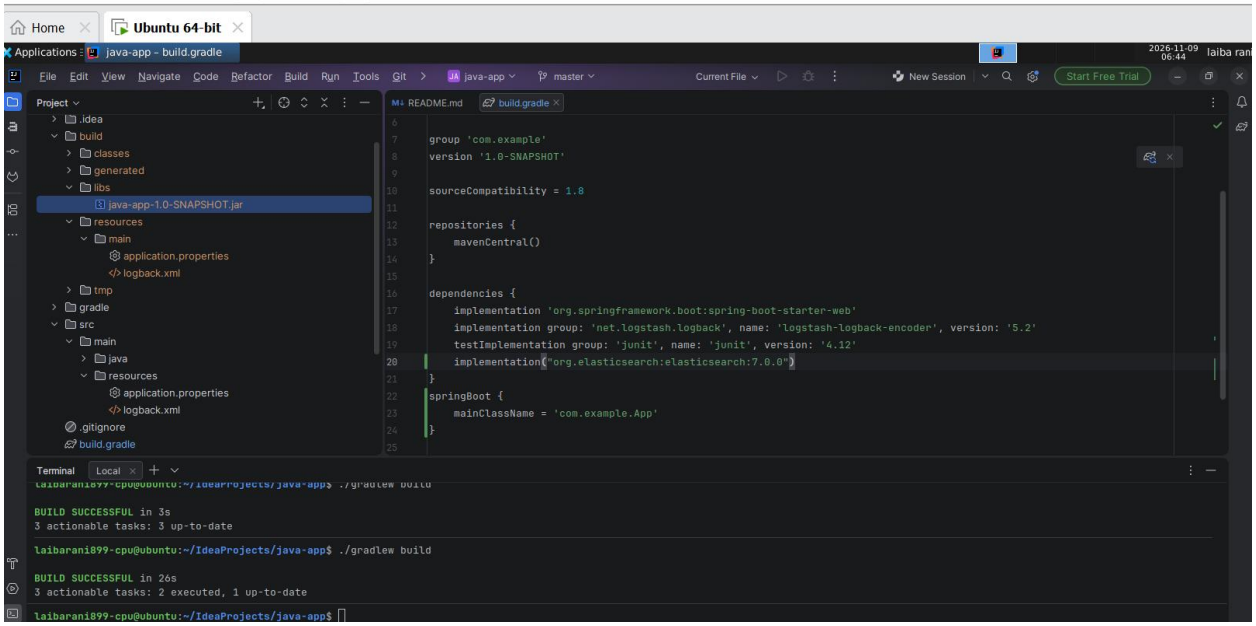
Open **View → Tool Windows → Terminal**. From the Gradle project root, run



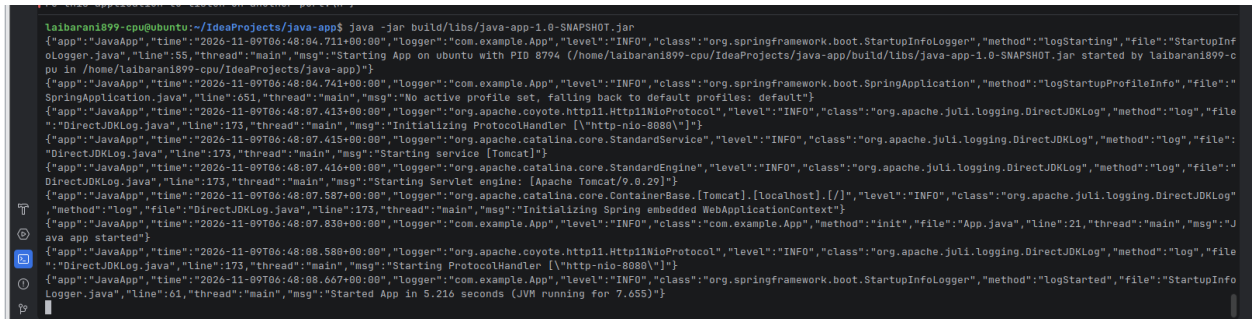
showing the dependency in `build.gradle` and the synchronized Elasticsearch library. Keep your required terminal prompt visible.



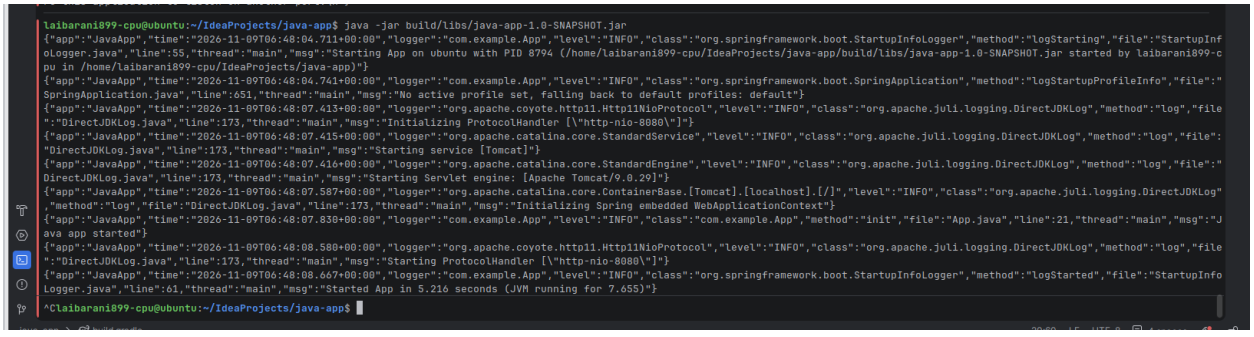
Build the project again after the dependency change



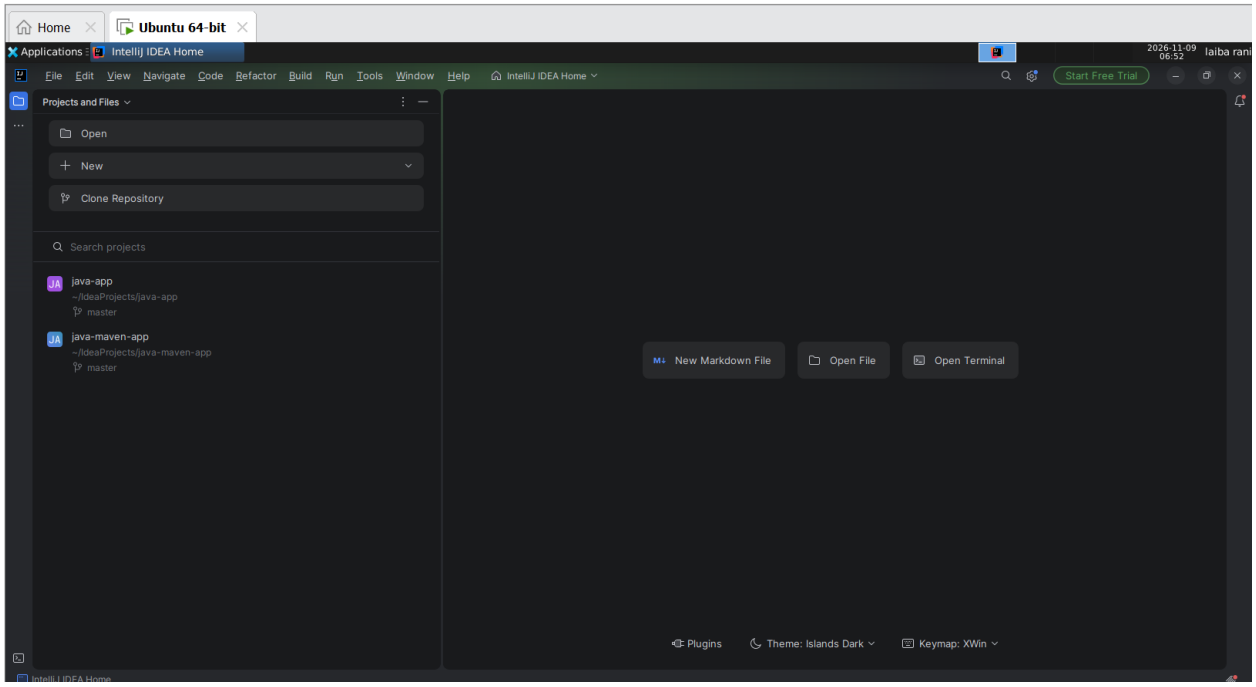
In IntelliJ's Project pane, expand **`build`** → **`libs`**.



Run the generated JAR from the project root:

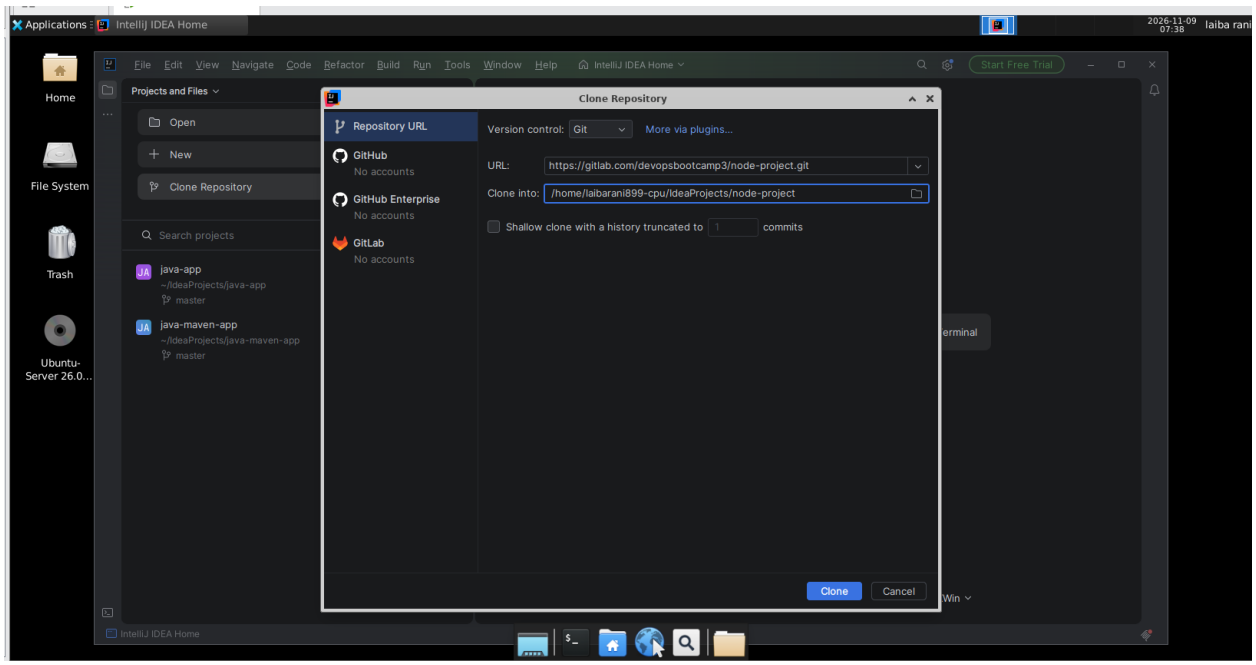


Press **Ctrl+C** and wait for the prompt to return. This also releases port 8080 before another application uses it.

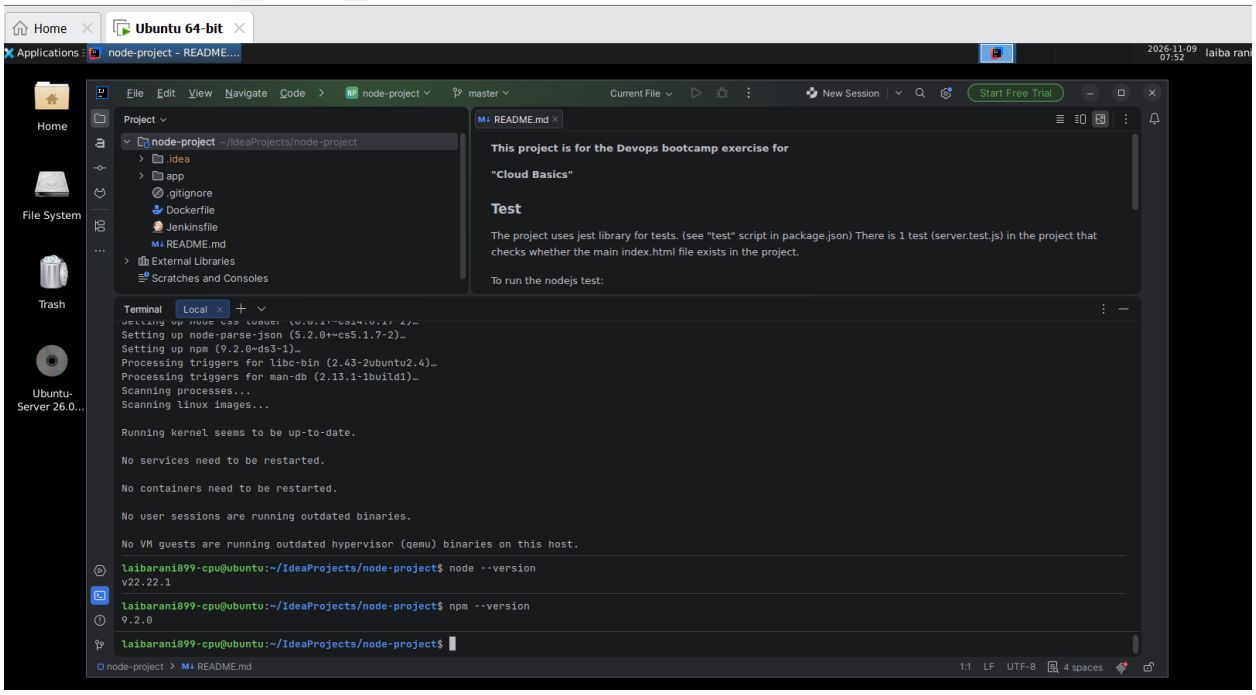


Select **File → Close Project**.

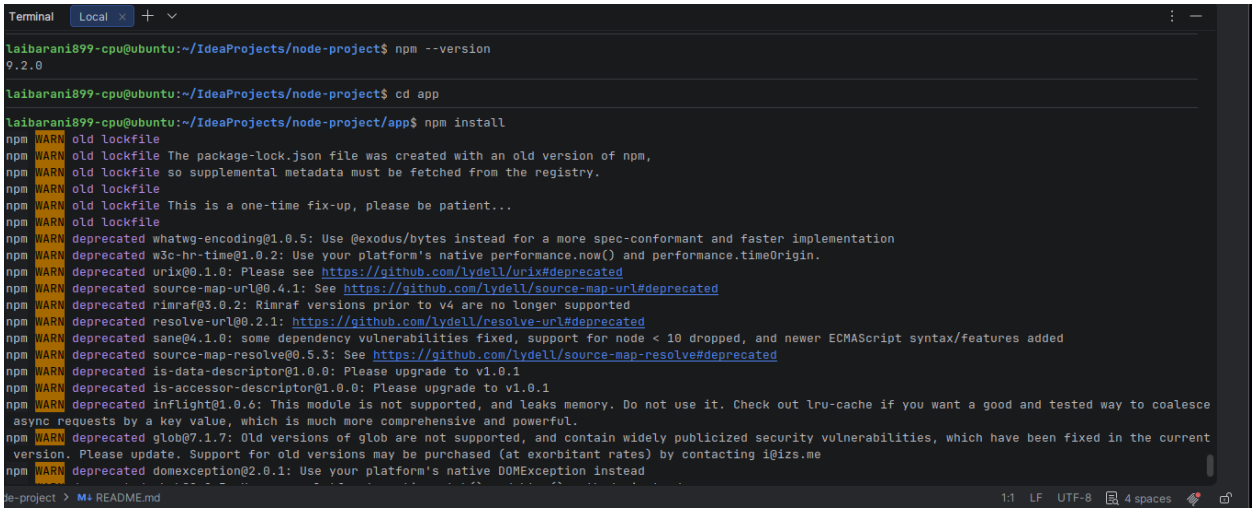
Task 4: Install and Run the Node.js Project



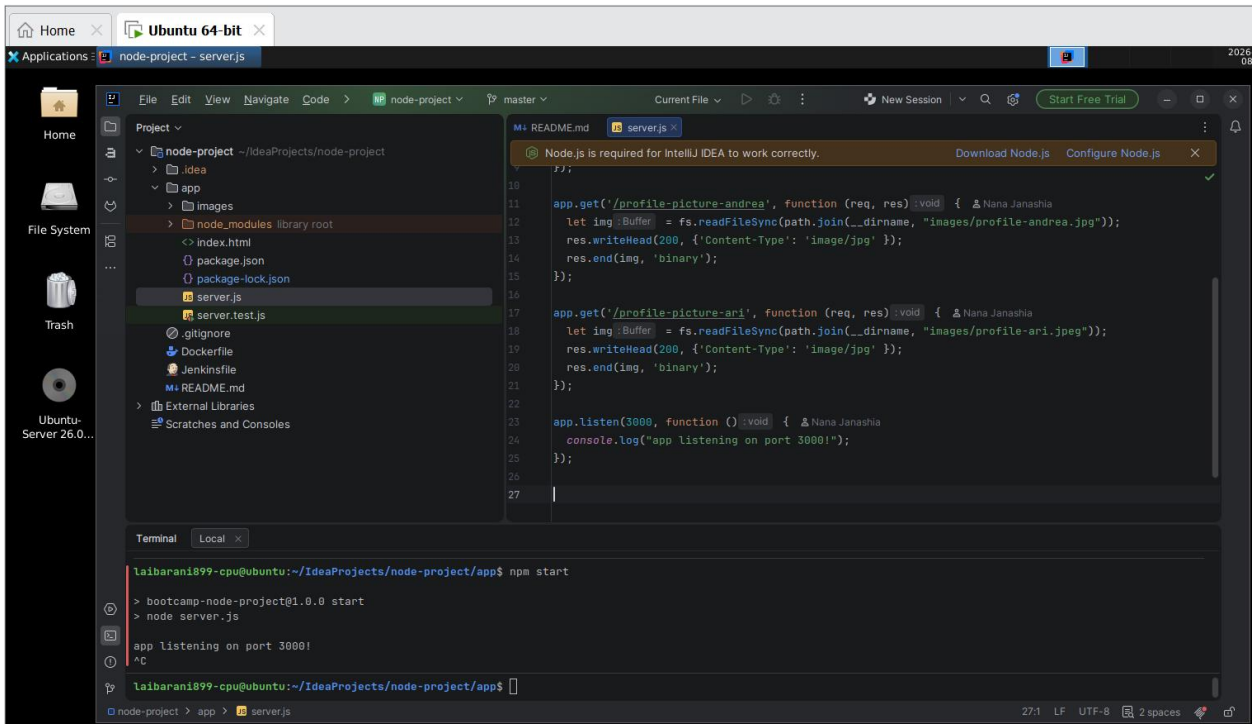
Use IntelliJ's **Clone Repository / Get from Version Control** action



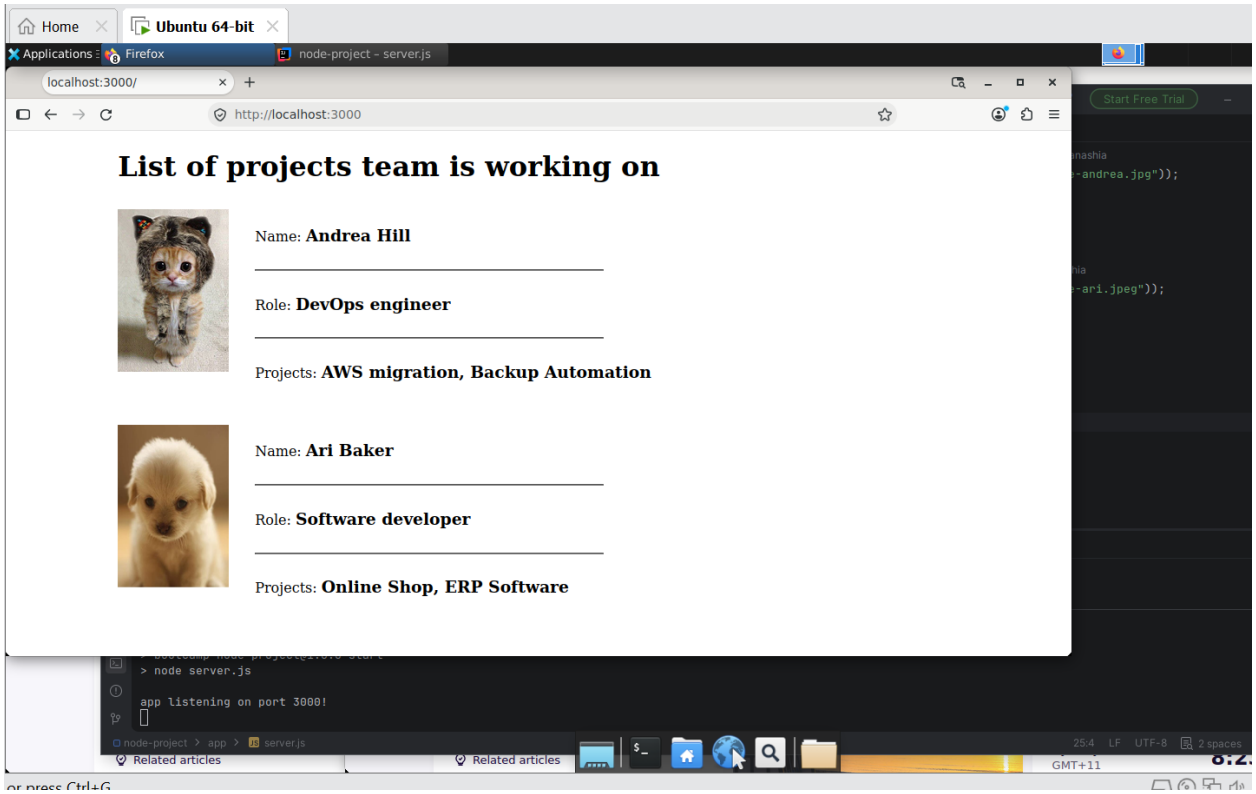
Open the terminal **inside IntelliJ** and install Node.js together with npm



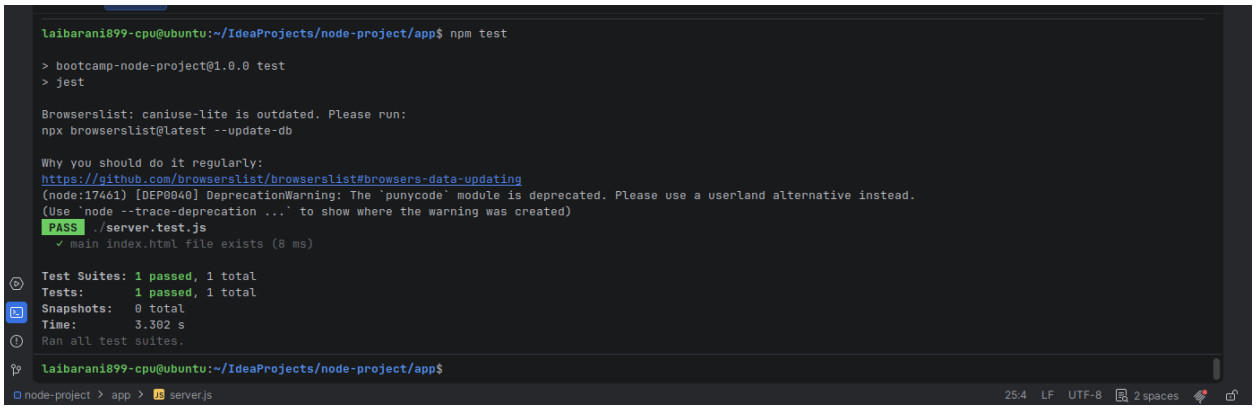
From the project root, enter `app` and install its dependencies



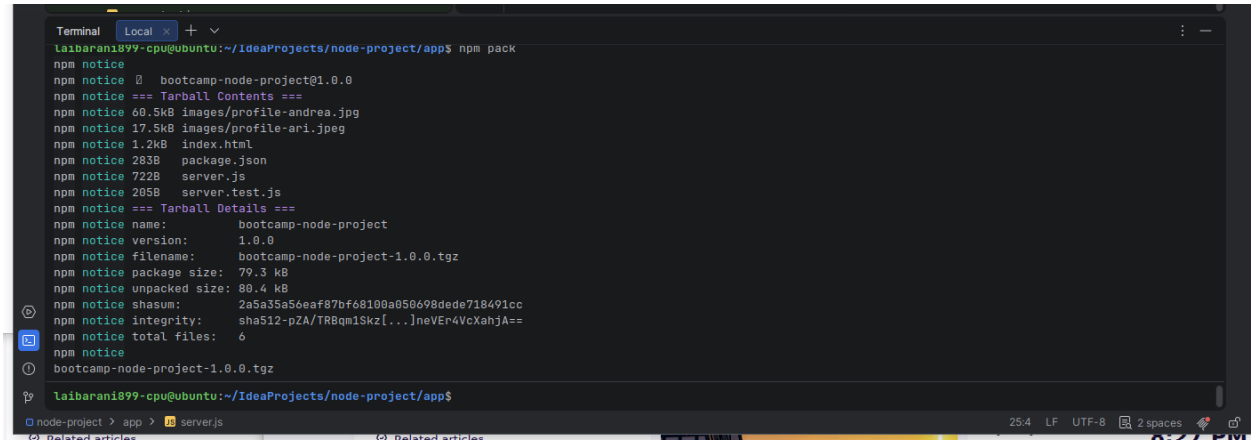
Start the application



Verify that the Node.js project page and its profile images are visible.

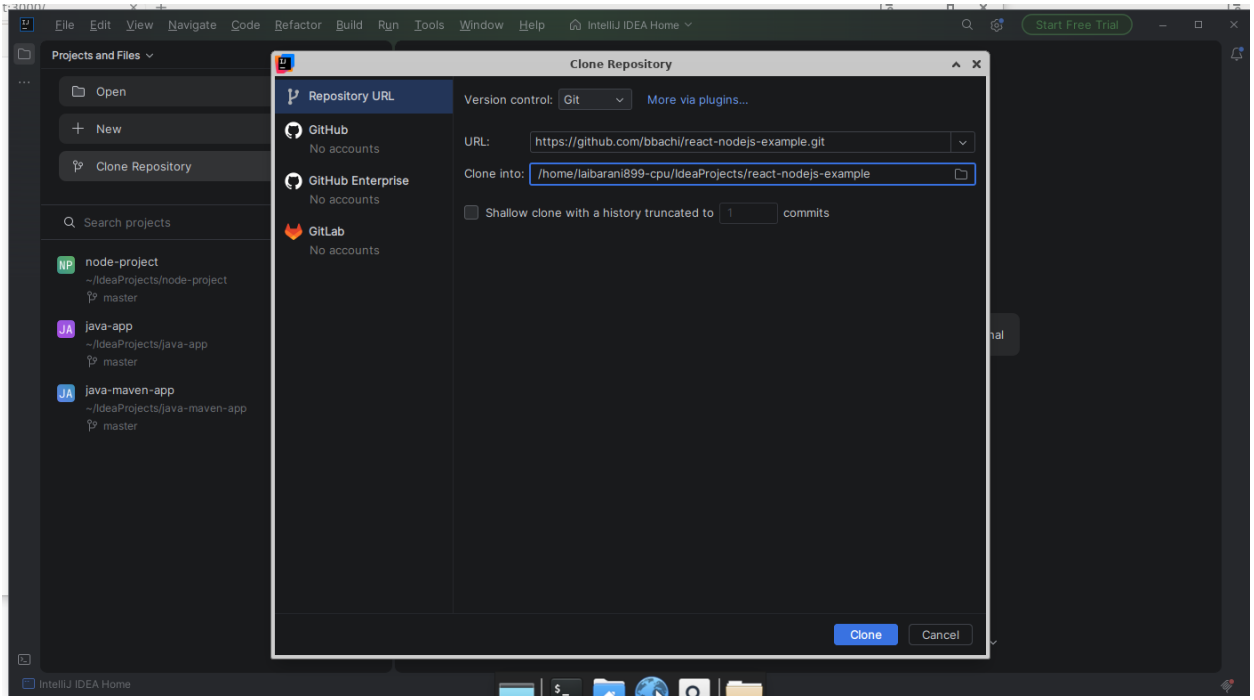


Stay in the **app** directory and run the application's tests

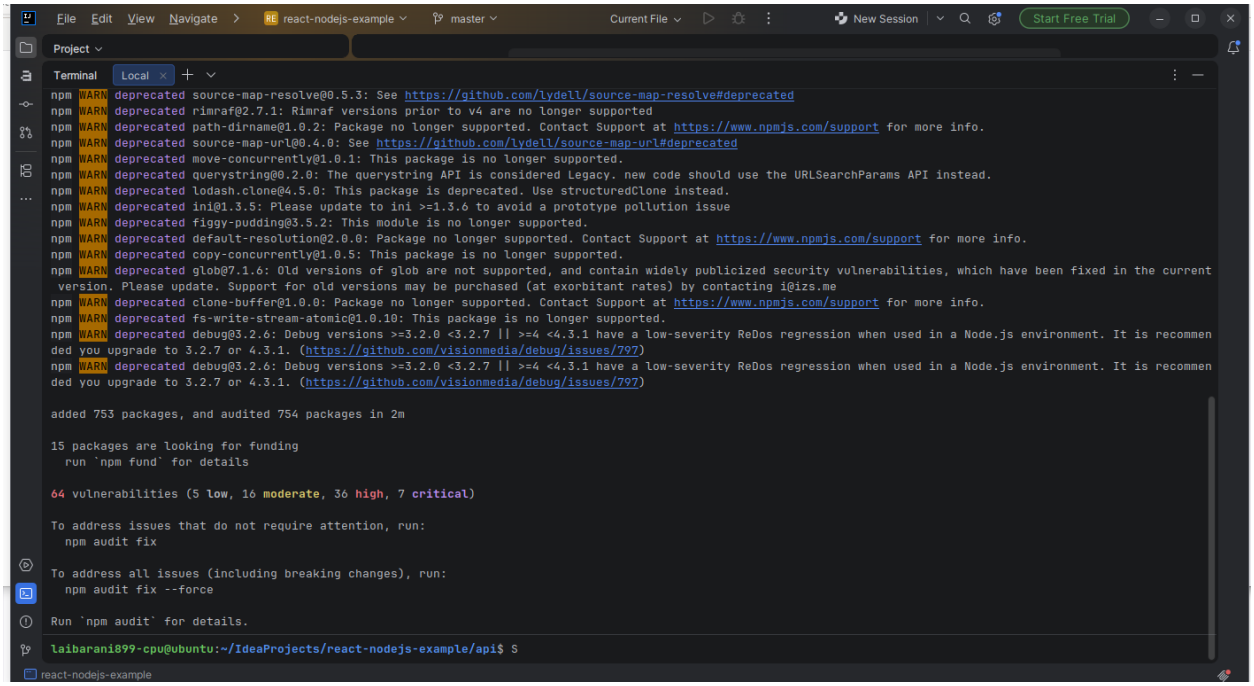


Create the application's package artifact

Task 5: Build the React and Node.js API Artifact



From IntelliJ's Welcome screen, use **Clone Repository**



Open the terminal **inside IntelliJ**. From the project root, run

```
laibarani899-cpu@ubuntu:~/IdeaProjects/react-nodejs-example/api$ npm run build

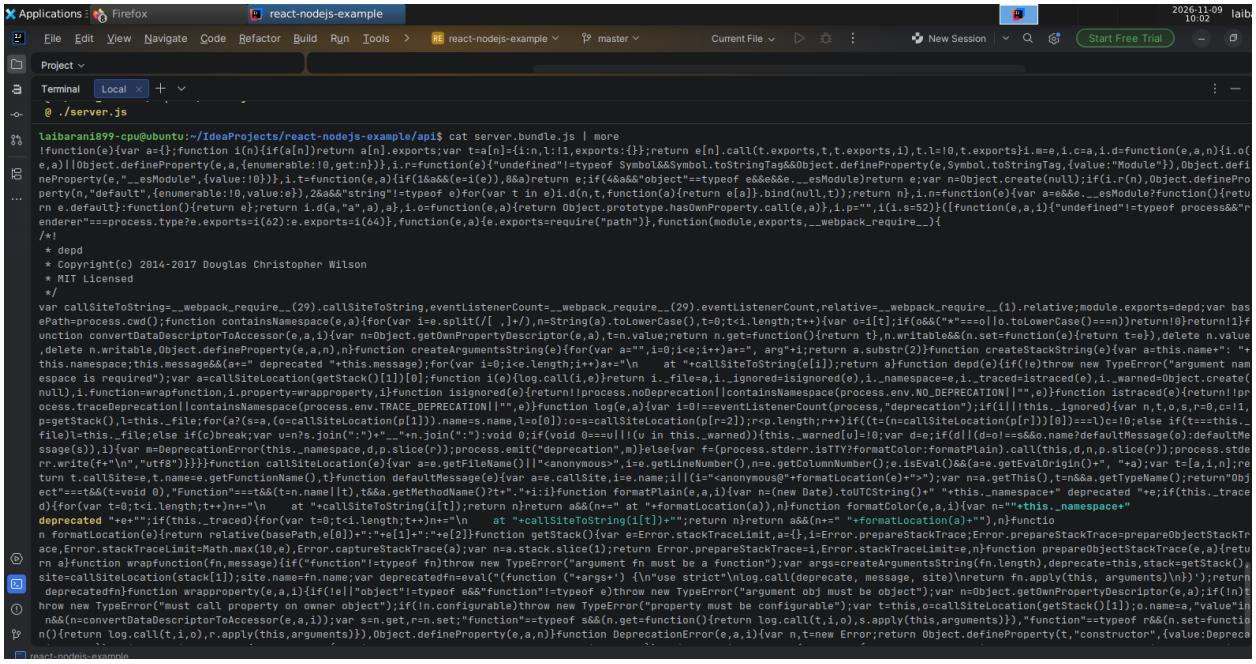
> react-nodejs-example@1.0.0 build

Structure Alt+7
Hash: abef9fc2cbaaf24f1946
Version: webpack 4.43.0
Time: 5672ms
Built at: 11/09/2026 9:59:04 AM
          Asset     Size  Chunks             Chunk Names
server.bundle.js  563 KiB       0  [emitted]  main
Entrypoint main = server.bundle.js
  [1] external "path" 42 bytes {0} [built]
  [4] external "fs" 42 bytes {0} [built]
  [6] external "buffer" 42 bytes {0} [built]
  [9] external "stream" 42 bytes {0} [built]
 [20] external "http" 42 bytes {0} [built]
 [22] external "util" 42 bytes {0} [built]
 [30] external "events" 42 bytes {0} [built]
 [32] external "net" 42 bytes {0} [built]
 [40] external "querystring" 42 bytes {0} [built]
 [41] external "url" 42 bytes {0} [built]
 [48] external "crypto" 42 bytes {0} [built]
 [52] ./server.js 990 bytes {0} [built]
 [65] external "tty" 42 bytes {0} [built]
 [86] external "zlib" 42 bytes {0} [built]
[101] ./node_modules/express/lib sync 160 bytes {0} [built]
    + 106 hidden modules

WARNING in ./node_modules/express/lib/view.js 81:13-25
Critical dependency: the request of a dependency is an expression
 @ ./node_modules/express/lib/application.js
 @ ./node_modules/express/lib/express.js
 @ ./node_modules/express/index.js
 @ ./server.js

laibarani899-cpu@ubuntu:~/IdeaProjects/react-nodejs-example/api$
```

In the same `api` directory, build the API artifact



View the generated bundle

Task 6: Build a Docker Image for the Node.js Project

```
laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1

laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$
```

Check that the Docker command is installed

```
laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$ ls
Dockerfile  Jenkinsfile  README.md  app

laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$
```

List the repository files and locate **`Dockerfile`**

```
laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$ sudo docker images
[sudo: authenticate] Password:
                                                                                                    i Info →  U In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest f79860ea5fac       11.8GB         3.06GB    U
mongo:8.2                            e0ce8c35124d        1.3GB          339MB    U
mysql:8.4                            0f51e63e537a       1.12GB          254MB    U

laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$
```

View the existing Docker images before building

```
laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$ sudo docker build -t lab5:1.0 .
[+] Building 349.4s (10/10) FINISHED                                                                                docker:default
 => [internal] load build definition from Dockerfile                                                                          0.2s
 => => transferring dockerfile: 156B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/node:13-alpine                                                             6.3s
 => [internal] load .dockerignore                                                                                             0.1s
 => => transferring context: 2B                                                                                               0.0s
 => [internal] load build context                                                                                            17.4s
 => => transferring context: 37.92MB                                                                                         17.2s
 => [1/5] FROM docker.io/library/node:13-alpine@sha256:527c70f74817f6f6b5853588c28de33459178ab72421f1fb7b63a281ab670258      42.5s
 => => resolve docker.io/library/node:13-alpine@sha256:527c70f74817f6f6b5853588c28de33459178ab72421f1fb7b63a281ab670258       0.1s
 => => sha256:4b9536424fa1675675e7edd59ae9a92573f0c010c5458733a493e2c3e1daf1ae 283B / 283B                                    0.6s
 => => sha256:780514bed1adda642a7f7008ea5b1477bb78a23e3f76eabbdde21031b2185a2f 35.30MB / 35.30MB                             35.0s
 => => sha256:5d74fb112a7d313344f9f679ccb975062377464235fac36e1b952aa2b125d5e2 2.24MB / 2.24MB                               5.3s
 => => sha256:cbdbe7a5bc2a134ca8ec91be58565ec07d037386d1f1d8385412d224deafca08 2.81MB / 2.81MB                               9.8s
 => => extracting sha256:cbdbe7a5bc2a134ca8ec91be58565ec07d037386d1f1d8385412d224deafca08                                    1.0s
 => => extracting sha256:780514bed1adda642a7f7008ea5b1477bb78a23e3f76eabbdde21031b2185a2f                                    6.9s
 => => extracting sha256:5d74fb112a7d313344f9f679ccb975062377464235fac36e1b952aa2b125d5e2                                    0.2s
 => => extracting sha256:4b9536424fa1675675e7edd59ae9a92573f0c010c5458733a493e2c3e1daf1ae                                    0.0s
 => [2/5] COPY app/ /usr/app/                                                                                                 7.7s
 => [3/5] WORKDIR /usr/app                                                                                                    0.3s
 => [4/5] RUN npm install                                                                                                   244.1s
 => [5/5] RUN npm run test                                                                                                    9.5s
 => exporting to image                                                                                                       37.9s
 => => exporting layers                                                                                                      27.2s
 => => exporting manifest sha256:670084d5c3a4b6c0b36332fec9654477bd36894934f5081050fc46a010bab607                            0.0s
 => => exporting config sha256:874aa557d5f8a7328fc0861d75f7bc8e62a930ebe9f4d3d2bdade5f4cb0c1507                              0.0s
 => => exporting attestation manifest sha256:cbdada22a37265a1894e05c4788462a7807ba37fd4c1ebf9ad79bf55a2f4957f                0.0s
 => => exporting manifest list sha256:f0167077ef7e982f22019e3e6371262de1724cd4dfd11d819f8bbc58f2044555                       0.0s
 => => naming to docker.io/library/lab5:1.0                                                                                   0.0s
 => => unpacking to docker.io/library/lab5:1.0                                                                               10.5s

laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$
```

Build the application image

```
laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$ sudo docker images
                                                                                                    i Info →  U In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest f79860ea5fac       11.8GB         3.06GB    U
lab5:1.0                             f0167077ef7e        417MB          85.2MB
mongo:8.2                            e0ce8c35124d        1.3GB          339MB    U
mysql:8.4                            0f51e63e537a       1.12GB          254MB    U

laibarani899-cpu@ubuntu:~/IdeaProjects/node-project$
```

List the images again